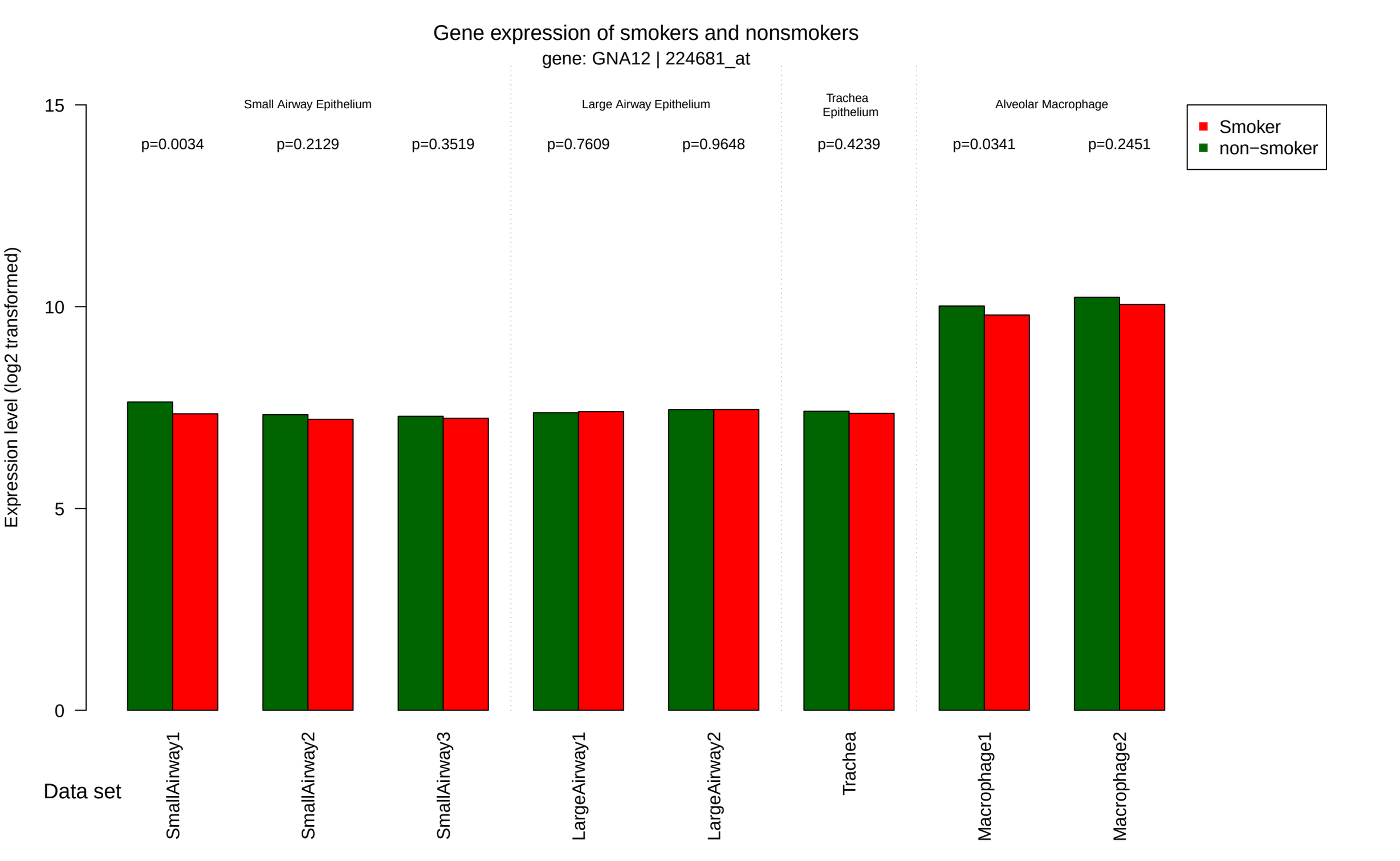
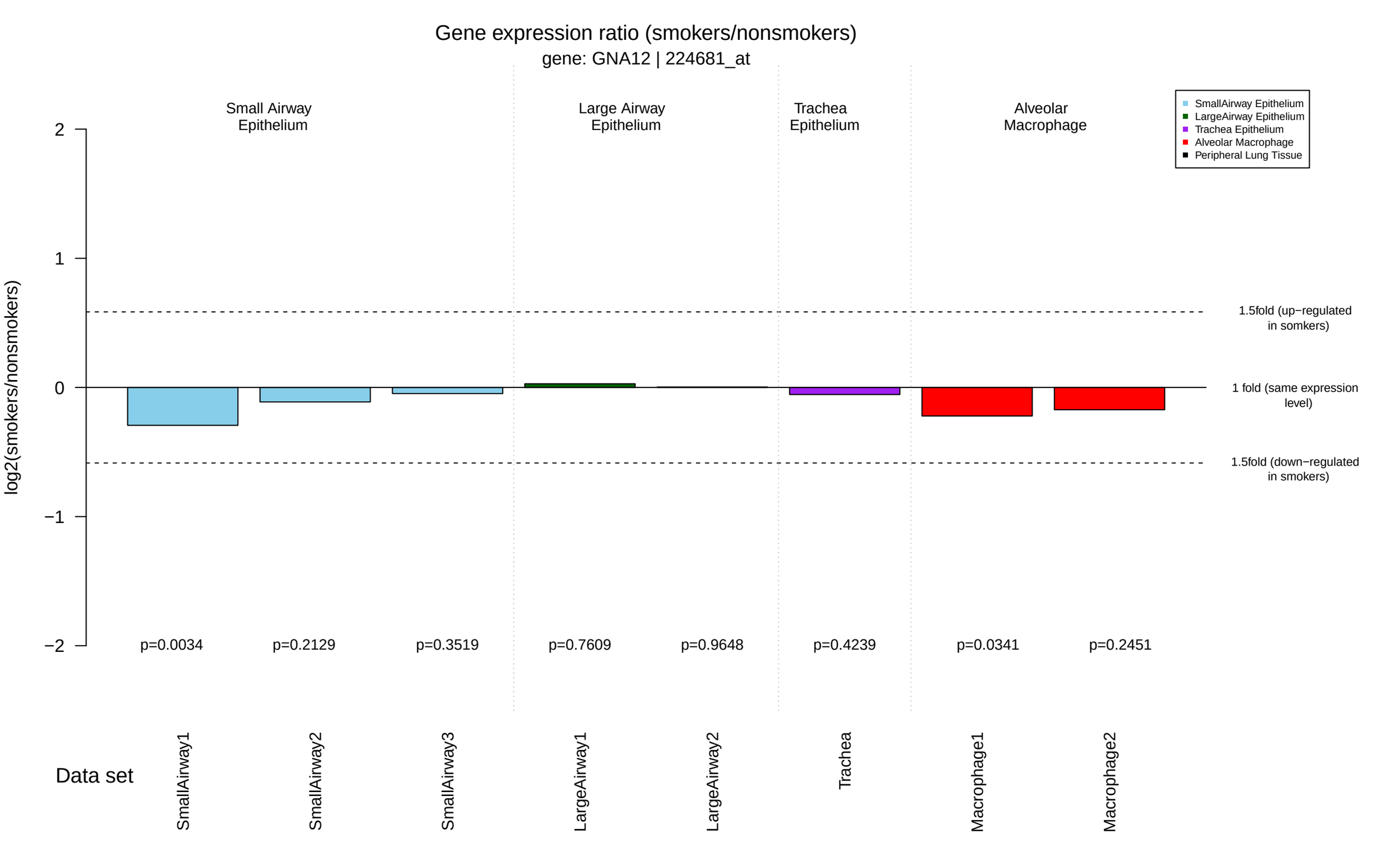
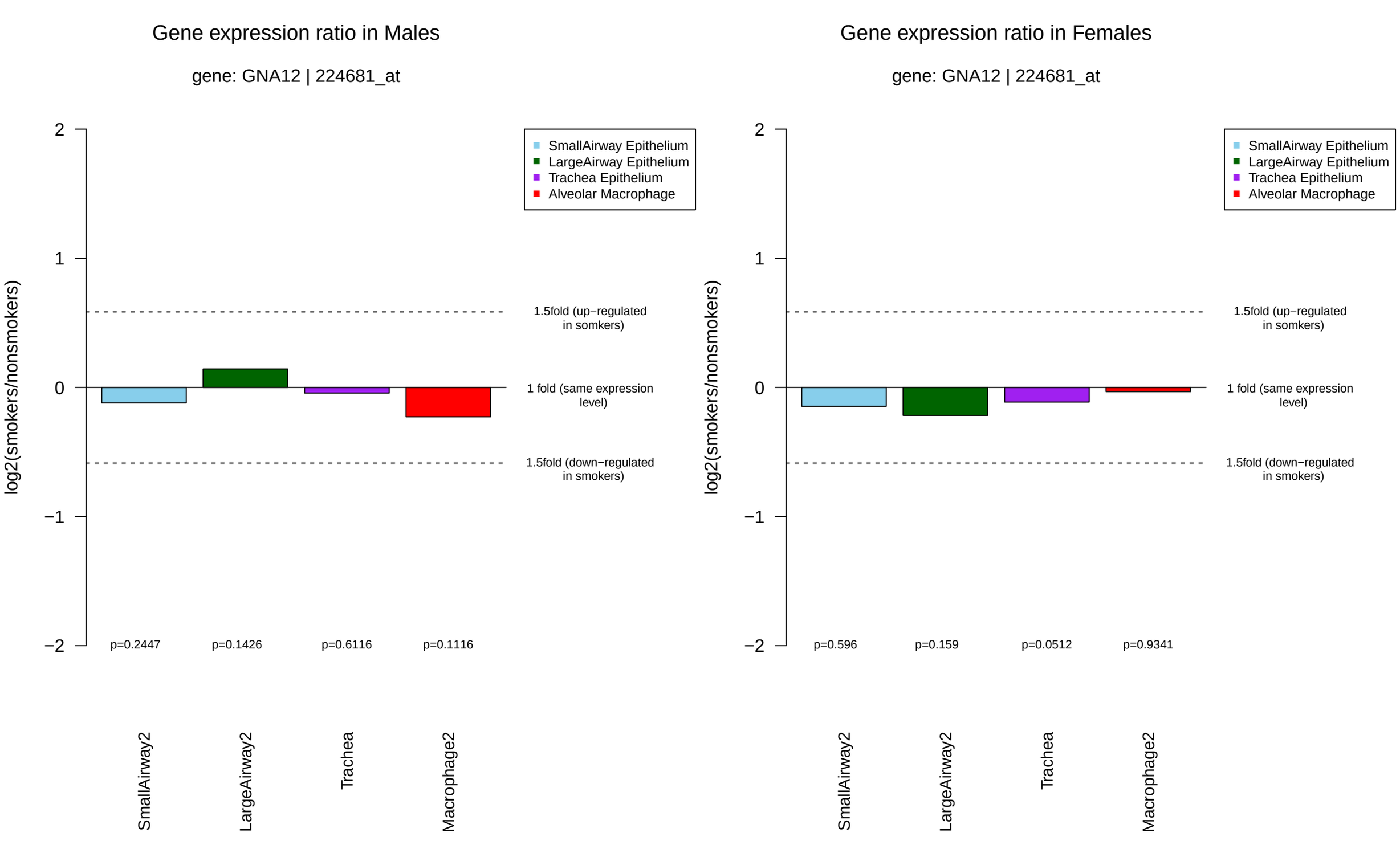
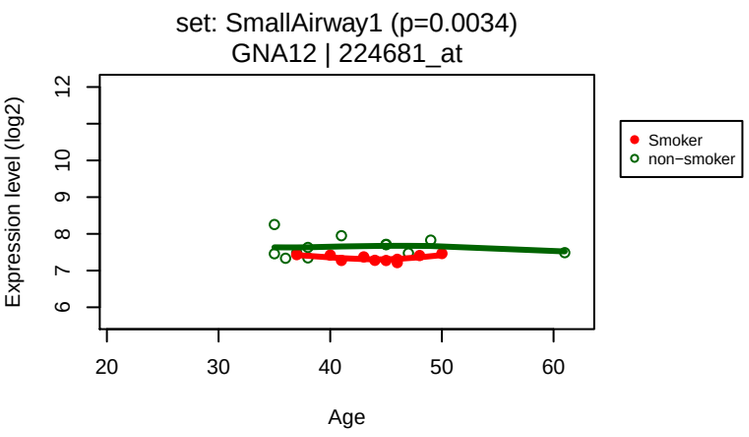
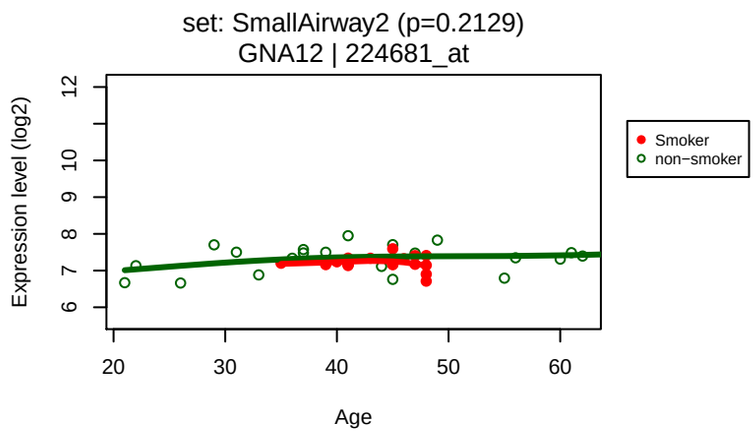
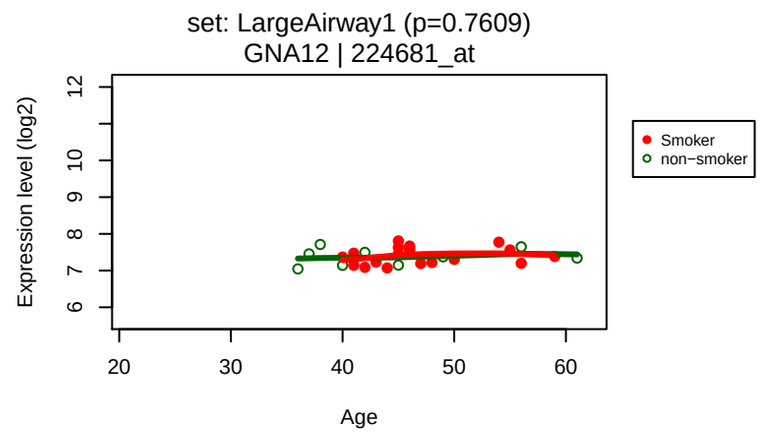

Gene expression of smokers and nonsmokers
gene: GNA12 | 224681_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0034
p=0.2129
p=0.3519
p=0.7609
p=0.9648
p=0.4239
p=0.0341
p=0.2451
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

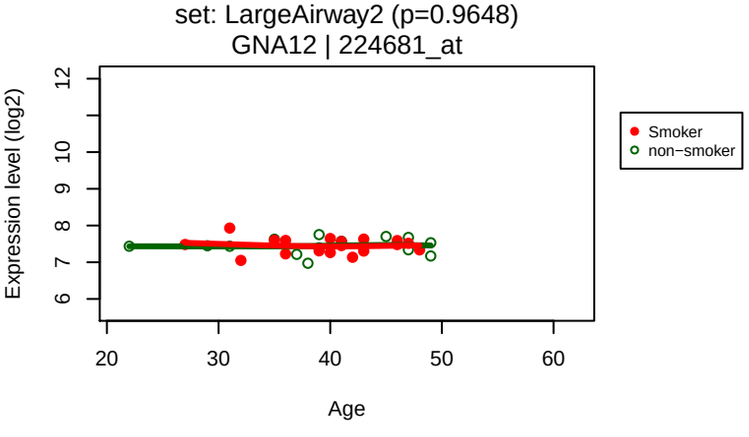

Gene expression ratio (smokers/nonsmokers)
gene: GNA12 | 224681_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0034
p=0.2129
p=0.3519
p=0.7609
p=0.9648
p=0.4239
p=0.0341
p=0.2451
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

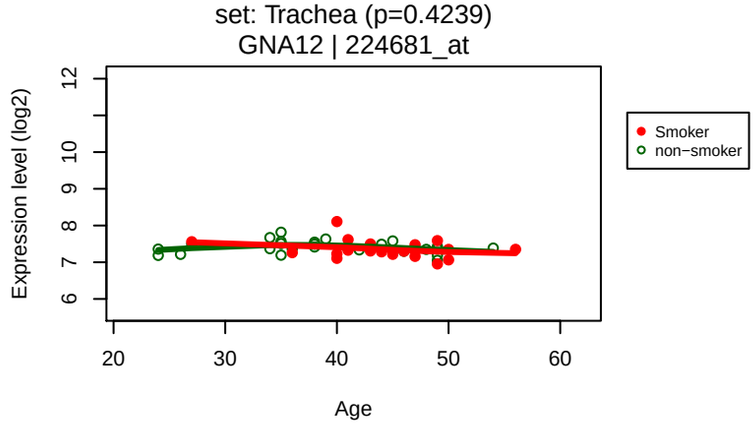

Gene expression ratio in Males
gene: GNA12 | 224681_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2447
p=0.1426
p=0.6116
p=0.1116
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNA12 | 224681_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.596
p=0.159
p=0.0512
p=0.9341
SmallAirway2
LargeAirway2
Trachea
Macrophage2

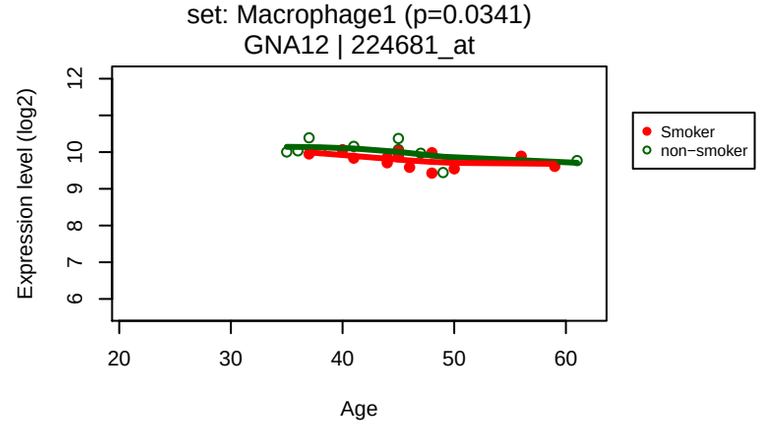

set: SmallAirway1 (p=0.0034)
GNA12 | 224681_at
Expression level (log2)
Age
Smoker
non−smoker

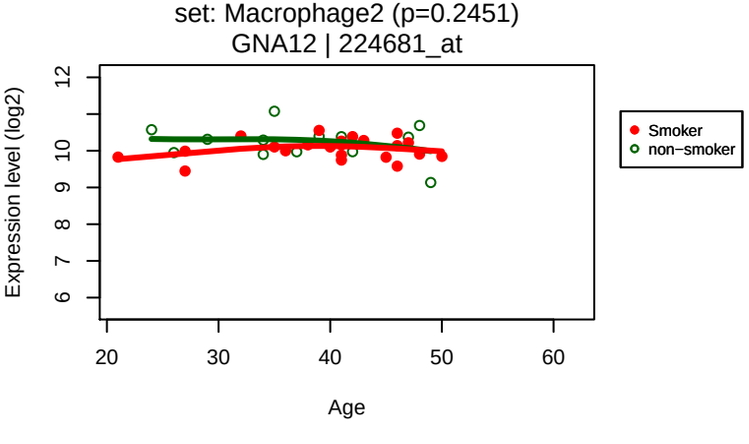

set: SmallAirway2 (p=0.2129)
GNA12 | 224681_at
Expression level (log2)
Age
Smoker
non−smoker

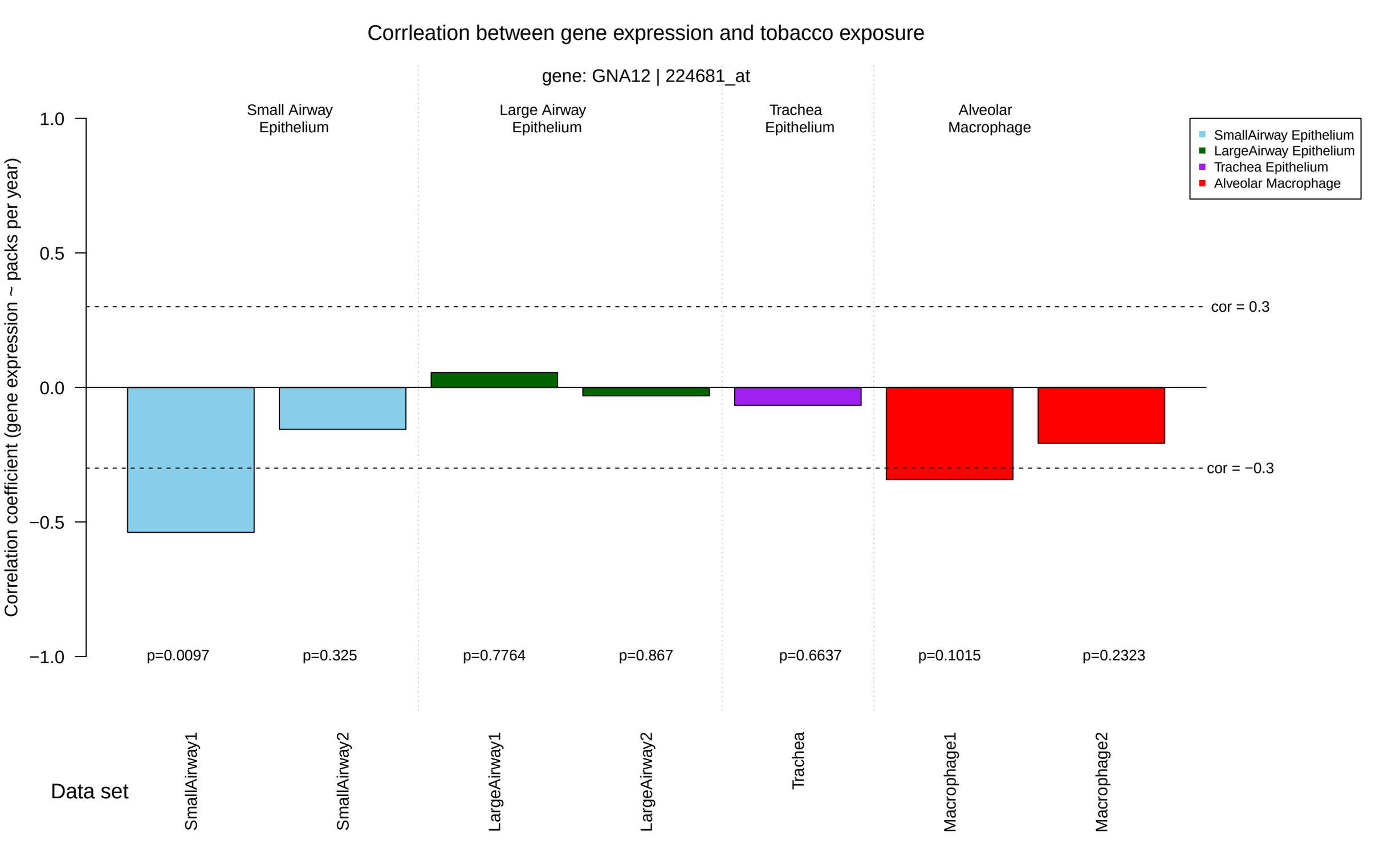

set: LargeAirway1 (p=0.7609)
GNA12 | 224681_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9648)
GNA12 | 224681_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4239)
GNA12 | 224681_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0341)
GNA12 | 224681_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2451)
GNA12 | 224681_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNA12 | 224681_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0097
p=0.325
p=0.7764
p=0.867
p=0.6637
p=0.1015
p=0.2323
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2